

1. Administrative & Study Identification

Full Study Title: Safety and Tolerability Study of XXB750 in Heart Failure Participants With Reduced or Mildly Reduced Ejection Fraction (HFrEF/HFmrEF) **Short Title/Acronym:** XXB750 Safety and Tolerability in HFrEF/HFmrEF **Protocol Number:** CXXB750A12101 **NCT Number:** NCT05328752 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** XXB750 (natriuretic peptide receptor A agonist) **Phase of Development:** Phase 1 **Medical Monitor:** Novartis Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of XXB750 in adult participants with chronic stable heart failure with reduced or mildly reduced ejection fraction (HFrEF/HFmrEF). **Secondary Objectives:** To evaluate the pharmacokinetics (PK) parameters of XXB750 (including AUCinf, AUClast, Cmax, and Tmax).

2.2 Background & Rationale To address the need for novel therapies in heart failure management, XXB750 acts as a natriuretic peptide receptor A agonist. This Phase 1 study is designed to assess whether single and multiple subcutaneous doses of XXB750 are safe, well-tolerated, and have a favorable pharmacokinetic profile when administered to patients with stable HFrEF/HFmrEF already receiving standard-of-care medications (such as ACEi/ARB, beta-blockers, or sacubitril/valsartan).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Sponsor open-label, participant- and investigator-blinded) **Randomization:** Randomized (2:1 ratio for Cohort 1; 3:1 ratio for Cohort 2)

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 24 participants (12 in Cohort 1, 12 in Cohort 2) **Number of Sites:** Multi-center **Estimated Study Start:** 17-May-2022 **Estimated Primary Completion:** 18-Jan-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Men or women age 18 to 80 years. 2. NYHA functional class II-III. 3. LVEF $\leq 50\%$ documented at

screening. 4. Systolic blood pressure 110-160 mmHg (Cohort 1) or 105-160 mmHg (Cohort 2), and heart rate 50-90 beats per minute, inclusive. 5. Treatment with a stable dose of a beta-blocker for at least 4 weeks. 6. *Cohort 1 only*: Stable therapies of ACEi/ARB for at least 4 weeks. *Cohort 2 only*: Stable dose of sacubitril/valsartan for at least 4 weeks.

4.2 Exclusion Criteria 1. Severe lung disease. 2. Recent acute heart failure or cardiovascular events. 3. Significant valve disease or recent device implantation in the heart. 4. Poor kidney function. 5. Severely overweight.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - *Cohort 1*: Single dose of XXB750 120 mg or placebo on Day 1. - *Cohort 2*: Three doses of XXB750 or placebo: Day 1 (120 mg), Day 28 (120 mg), and Day 56 (240 mg). **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 1 day of dosing for Cohort 1 (followed by 13 weeks observation); 56 days of dosing for Cohort 2 (followed by 13 weeks observation).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care heart failure medications, including beta-blockers, ACEi/ARBs, and sacubitril/valsartan. **Prohibited:** Investigational treatments or newly initiated (non-stable) HF medications not conforming to the background therapy criteria.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -29 to Day -1	Informed Consent, Medical History, LVEF confirmation, Vitals
Baseline/ Domicile	Day -2 to Day -1	Admission to clinical facility for baseline safety assessments
Cohort 1 Dosing	Day 1 to Day 5	Single SC dose, 5-day domiciling period, safety & PK sampling
Cohort 2 Dosing	Days 1, 28, 56	Three SC doses; re-domiciled for 3 days around each dose (Days 28-30, 56-58)
End of Study	Day 91 (C1) / Day 146 (C2)	Final safety, PK, and tolerability assessments out to 13 weeks post-dose

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Number and percentage of participants with treatment-emergent adverse events, and changes in vital signs, ECGs, and laboratory test results to evaluate safety and tolerability. **Measurement Method:** Clinical observation, continuous ECG monitoring, and central laboratory analysis during domiciling and follow-up visits.

7.2 Secondary & Exploratory Endpoints - Pharmacokinetics of XXB750: Area under the plasma concentration curve (AUCinf, AUClast). - Pharmacokinetics of XXB750: Peak plasma concentration (Cmax) and time to maximum concentration (Tmax).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Conducted continuously during in-patient domiciled periods and throughout the 13-week post-dosing follow-up. **Serious Adverse Events (SAEs):** Standard reporting timelines applied. Stepwise safety measures included sentinel dosing of the first two participants in Cohort 1 (one active, one placebo). **Data Monitoring Committee (DMC):** Internal safety monitoring committee managed by Novartis to oversee dose escalations.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: - *Safety Analysis Set:* All participants who received at least one dose of study treatment. - *PK Analysis Set:* All participants with at least one available valid PK concentration measurement who received XXB750. **Sample Size Justification:** A total of 24 subjects (12 per cohort) provides sufficient data for preliminary Phase 1 safety, tolerability, and PK parameter estimation. **Handling of Missing Data:** Standard Phase 1 protocols apply; participants analyzed based on available PK points.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring due to Phase 1 domiciling requirements and PK sample integrity checks. **Audit Trail:** Electronic case report forms (eCRF) utilized for all vital signs and safety labs, with strict audit trails for AE and SAE resolutions.

11. Study Identifiers & Governance

Primary ID: NCT05328752 **Registry Number:** NCT05328752 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Safety and Tolerability Study of XXB750 in Heart Failure Participants With Reduced or Mildly Reduced Ejection Fraction (HFrEF/HFmrEF) **Official Regulatory Title:** A multi-center, randomized, sponsor open-label, participant- and investigator-blinded, placebo-controlled, single and multiple dose study to investigate the safety and tolerability of XXB750 in HFrEF/HFmrEF

12. Clinical Framework (Hyperkalemia Specific)

(Note: Framework adapted to Heart Failure context as this study does not investigate Hyperkalemia) **Primary Condition:** Heart Failure with Reduced or Mildly Reduced Ejection Fraction (HFrEF/HFmrEF) **Diagnosis Threshold:** LVEF $\leq 50\%$, NYHA Class II-III **Medical Methodology:** Subcutaneous administration of Natriuretic Peptide Receptor A Agonist (XXB750) **Optimization Target:** PK exposure and safety/tolerability confirmation

13. Study Procedures & Methodology

Management Pathway: Phase 1 Domiciled Safety & PK Observation **Education Protocol (HCP):** Investigator training on SC administration and AE tracking **Education Protocol (Patient):** In-patient instruction during domiciling periods **Quality Audit Mechanism:** Source data verification and PK sample reconciliation

14. Observation & Follow-Up Schedule

Total Duration: Up to 146 days (13 weeks post final dose) **Onsite Visit Cadence:** Screening, Day -2 to Day 5 domicile, plus periodic follow-ups to Day 91 (C1) / Day 146 (C2) **Remote Monitoring Frequency:** N/A - Intensive onsite Phase 1 monitoring **Checklist Review Interval:** Daily during domiciling

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	03-Apr-2026				
Clinical Operations Lead	[Name]	_____	03-Apr-2026		Lead		
Statistician	[Name]	_____	03-Apr-2026				